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Experiment 2

Area of a Circle and Stages of C Compilation

We will write and execute a C program in the Kali Linux terminal using the cc compiler to calculate the area of a circle and study the different stages of compilation.

Formula of area of circle:

The area of a circle is calculated from its radius= πr^2

where r is the radius and $(\pi) \approx 3.14$.

1. Create the C program :

Open the Kali Linux terminal and create a source file- " vi circle.c " , now the command will change into interim mode . Press " i " to go to the insert mode , write the c program in insert mode and press " esc " key to go back to the interim mode ,press " :wq " to go back to the command mode again

Write the following program:

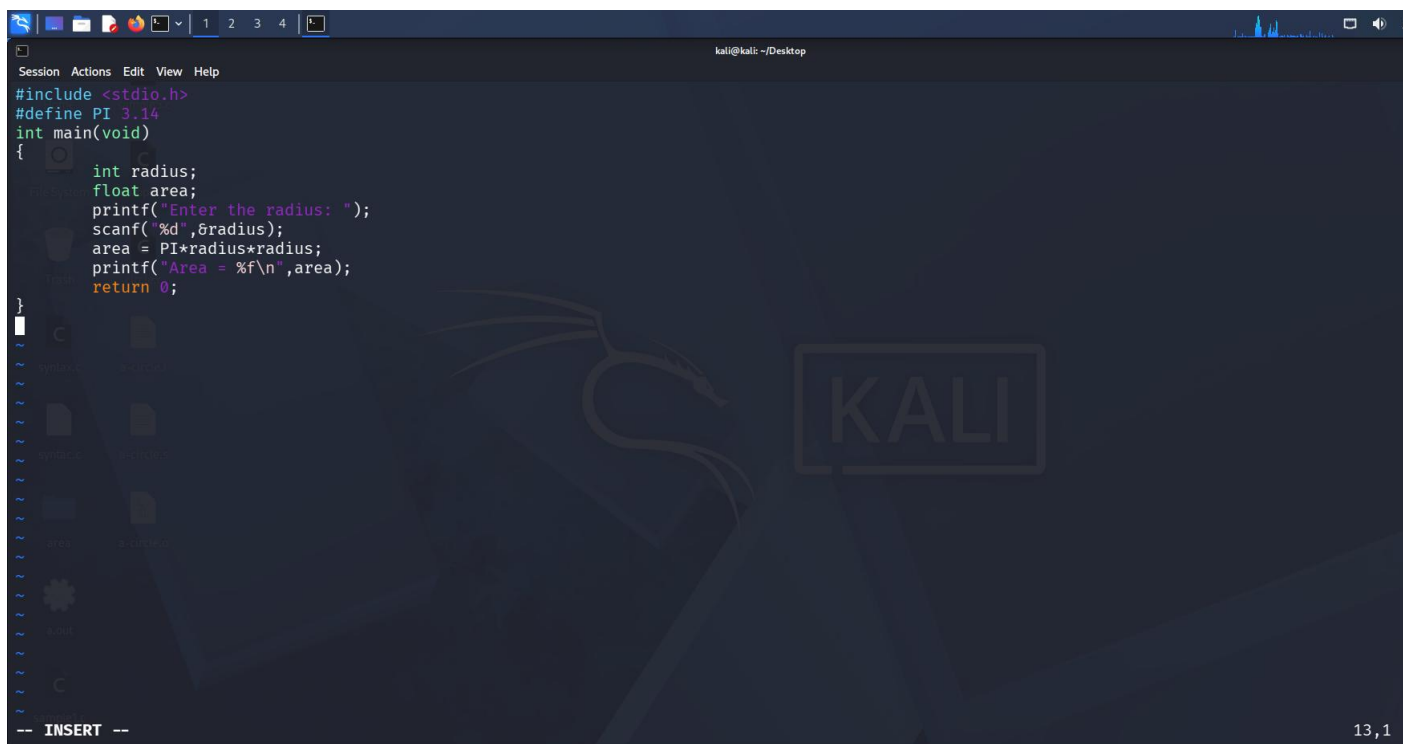
```
#include <stdio.h>
```

```
#define PI 3.14
```

```
int main(void)
```

```
{  
  
    int radius;  
  
    float area;  
  
    printf("Enter the radius: ");  
  
    scanf("%d", &radius);  
  
    area = PI * radius *radius;  
  
    printf("Area = %f\n", area);  
  
    return 0;  
}
```

Here the screenshot of the program in insert mode

A screenshot of a Kali Linux terminal window. The terminal has a dark blue background with a faint Kali Linux logo and the word "KALI" in a box. The window title is "kali@kali: ~/Desktop". The menu bar shows "Session Actions Edit View Help". The code being edited is a C program for calculating the area of a circle. The code is as follows:

```
#include <stdio.h>  
#define PI 3.14  
int main(void)  
{  
    int radius;  
    float area;  
    printf("Enter the radius: ");  
    scanf("%d", &radius);  
    area = PI*radius*radius;  
    printf("Area = %f\n", area);  
    return 0;  
}
```

The cursor is at the end of the closing brace of the main function. At the bottom of the terminal, it says "-- INSERT --" and "13,1".

2. Compile while saving intermediate files

Run:

“ cc -save-temps circle.c “

The -save-temps option tells the compiler to keep the intermediate files created during compilation.

Check them using:

“ ls “

You will generally find:

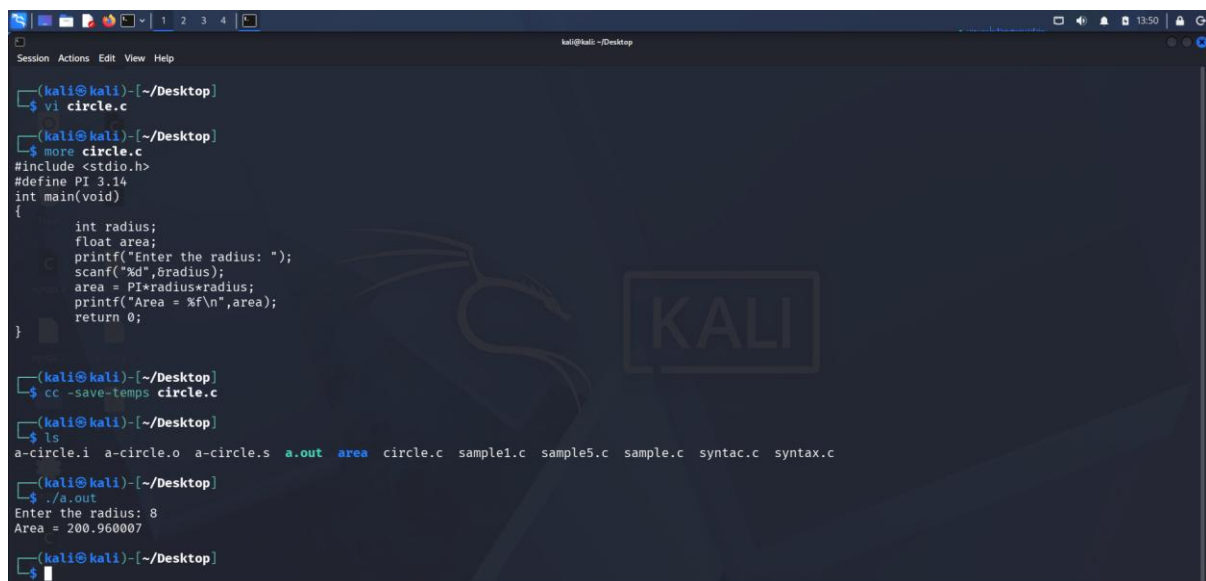
circle.c

circle.i

circle.s

circle.o

circle



```
(kali@kali)-[~/Desktop]
$ vi circle.c

(kali@kali)-[~/Desktop]
$ more circle.c
#include <stdio.h>
#define PI 3.14
int main(void)
{
    int radius;
    float area;
    printf("Enter the radius: ");
    scanf("%d",&radius);
    area = PI*radius*radius;
    printf("Area = %f\n",area);
    return 0;
}

(kali@kali)-[~/Desktop]
$ cc -save-temps circle.c

(kali@kali)-[~/Desktop]
$ ls
a-circle.i  a-circle.o  a-circle.s  a.out  area  circle.c  sample1.c  sample5.c  sample.c  syntac.c  syntax.c

(kali@kali)-[~/Desktop]
$ ./a.out
Enter the radius: 8
Area = 200.960007

(kali@kali)-[~/Desktop]
$
```

3. Stages of Compilation

The complete compilation process is:

circle.c

|



Preprocessing

|



circle.i

|



Compilation

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circle.s

|



Assembly

|



circle.o

|



Linking

|



circle

|



Execution

Stage 1 — Preprocessing: The preprocessor processes statements beginning with #, such as:

```
#include <stdio.h>
```

It produces the preprocessed file circle.i.

Stage 2 — Compilation: The compiler converts the preprocessed C code into assembly language and produces circle.s.

Stage 3 — Assembly: The assembler converts assembly code into object/machine code, producing circle.o.

Stage 4 — Linking: The linker links circle.o with the required libraries. For example, library functions such as printf() and scanf() must be resolved. It creates the final executable:

4. Execute the Program

Run:

“ ./a.out “

Example:

Enter the radius of circle: 8

Area of circle = 200.960007

Result: The C program was successfully written, compiled using the cc compiler, and executed in the Kali Linux terminal. The stages of preprocessing, compilation, assembly, and linking were also observed using the intermediate files generated with -save-temps.

In short:

.c → .i → .s → .o → executable

Source → Preprocessor → Compiler → Assembler → Linker → Execution